



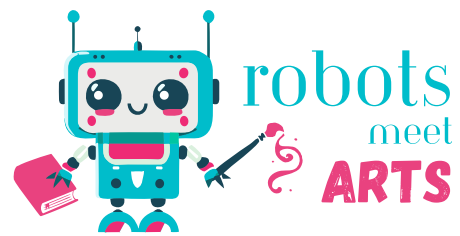
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000151881

THE COMPASS ROBOT

DIY Robotic Object for Getting Around





Subject: Geography

Content: Orienting oneself in space, navigation, reading a map, using a compass

Computational practices:

Decomposition, pattern recognition, abstraction, algorithms, problem-solving; block programming, sensor use, digital making

Unplugged/plugged:
plugged



Preparation time:

60 - 90 min



Age:

8-12 years



Duration: 3 classroom

sessions + 1 orienteering outing if possible



Location:

indoor/outdoor

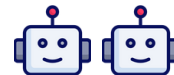


Group distribution:

full classroom/in group



Difficulty



This sequence allows pupils to develop their spatial orientation skills by building a concrete tool: a cardboard robot equipped with a digital compass programmed on micro:bit. By combining manual construction and programming, pupils discover how a compass works, understand the concepts of cardinal points, and learn to orient themselves in their immediate environment.

The following competencies are developed:

- **"Orienting oneself in space: building geographical reference points"** – Pupils learn to name and locate the cardinal points, orient a map, and find their way in a familiar space
- **"Reasoning, justifying an approach"** – Formulating hypotheses about how the magnetometer works and verifying them through experimentation
- **"Practising different languages in geography"** – Reading and orienting maps, using directional symbols, creating orienteering courses
- **"Cooperating and sharing"** – Collaborative construction of the robot and orienteering courses



Cross-Curricular Links

- **Science and Technology**: Pupils discover terrestrial magnetism and how a compass works. They practise a scientific approach by testing and calibrating their sensor. Programming introduces them to "designing and producing a technical object as a team."
- **Mathematics**: Work on angles (90° , 180° , 270° , 360°), directions, and geometry in space. Pupils manipulate angular measurements and understand the relationship between degrees and directions.
- **Physical Education**: Orienteering allows pupils to use orientation skills in a sporting and fun context, using the compass robot as an aid.
- **Art and Design**: Designing and decorating the cardboard robot develops creative and design skills.

Helpful links:

- <https://makecode.microbit.org/projects/compass> – MakeCode compass project
- <https://microbit.org/projects/make-it-code-it/compass-bearing/> – Compass bearing project
- <https://www.instructables.com/Microbit-Compass/> – Instructables micro:bit compass tutorial
- <https://www.youtube.com/watch?v=lg3VLcSRUNc> – Cardboard robot compass build
- <https://www.youtube.com/watch?v=9lyssvevhJ0> – How compasses work
- <https://www.youtube.com/watch?v=KeDDMq60stI> – micro:bit compass tutorial

Material Needed

- **Basic Materials**

- Documentation and Observation: traditional compasses (at least one for demonstration), maps of the neighbourhood or school, notebooks, pencils
- Robot Construction: thick cardboard or recycled boxes, scissors, glue, tape, felt-tip pens and paint for decoration
- Teaching Resources: images of cardinal points, compass diagrams, explanatory videos about magnetism

- **For the Robotic Compass**

- micro:bit Board: The micro:bit board includes a magnetometer (magnetic field sensor) that allows you to create a digital compass. It also has a 25-LED matrix perfect for displaying directions. Recommended complete kit (approximately 20 EUR): <https://www.kubii.com/fr/kits-micro-controleurs/3092-kit-microbit-go-v2-5051259252592.html>
- Power Supply: AAA battery pack to make the robot autonomous
- Programming: Access to a computer with MakeCode (free online: makecode.microbit.org)

- **For Orienteering Courses (Optional)**

- Coloured markers or cones to mark waypoints
- Printed maps of the course (schoolyard, nearby park)
- Stopwatches for timed challenges
- Logbooks for pupils to record directions and observations

Note for teachers: Not every pupil needs their own micro:bit board. Two or three boards are sufficient for the entire class. Pupils can take turns programming on the MakeCode emulator, then test on the physical boards. The cardboard robot can be designed with a modular slot to accommodate the shared board.

Preamble on the Robotic Orientation of the Project

Educational robotics is often associated with expensive kits, but the aim of this project is to demonstrate that with a simple micro:bit board, some recycled cardboard, and a bit of creativity, pupils can build a functional robot that teaches them to orient themselves in space.

For what is a robot, if not a physical object capable of perceiving its environment (here, the Earth's magnetic field) and responding to it (displaying a direction, rotating an arrow)? No need for articulated arms or motorised wheels: our cardboard compass robot perfectly embodies this link between code and the real world. The pupil programmes instructions, and a concrete object that they have built with their own hands responds to those instructions. It is this magic of seeing the digital act upon the physical that makes robotics captivating.

Here, pupils don't code for the sake of coding: they code to understand how a compass works, to orient themselves in their neighbourhood, to create orienteering courses. The robot is not an end in itself, but a learning companion that makes tangible abstract concepts such as cardinal points, angles, or terrestrial magnetism.

Finally, this project demonstrates that robotics can be done on a minimal budget. A micro:bit board shared between several groups, cereal boxes, glue: these are the ingredients of a robot that is every bit as good as commercial kits, with the added advantage of being entirely customisable and understandable by pupils.

Introduction – The Challenges of Spatial Orientation

Knowing how to orient oneself in space is a fundamental skill that humans have developed for millennia. From stars to magnetic compasses, from paper maps to GPS, orientation tools have evolved but the need remains the same: knowing where you are and which direction to go.

In this sequence, pupils will build their own orientation tool: a cardboard robot equipped with a digital compass. This project will enable them to understand concretely how a compass works, to manipulate the concepts of cardinal points and degrees, and to develop their spatial orientation skills.

The compass robot becomes a learning companion: its "face" (the LED matrix of the micro:bit board) displays an arrow pointing North, allowing pupils to orient themselves in a fun way. Building the cardboard robot adds a creative and customisable dimension to the project, whilst programming develops algorithmic thinking.

The final objective is twofold: mastering the spatial orientation concepts in the geography curriculum, whilst discovering the possibilities offered by programming and digital sensors.

Preparation and teacher notes

- Obtain one or more traditional compasses for demonstration
- Prepare "N", "S", "E", "W" signs to place at the four corners of the classroom
- Identify in advance where North is in your classroom (check with a compass or smartphone app)
- Select a short video on terrestrial magnetism suitable for your pupils
- Print a map of the neighbourhood or school for hands-on activities

Warm-up: Discovering Orientation and Compasses

Objective: Understand what orientation is, discover the cardinal points and how a compass works

Begin with an open discussion with the class: "What does it mean to orient oneself? When do you need to orient yourselves in everyday life?" Note the pupils' responses on the board. Draw out different situations: finding your way in an unfamiliar town, navigating in a forest, following a route on a map...

Then introduce the etymology of the word "orientation": it comes from "orient", which means the East, where the sun rises. Historically, humans found their bearings using the sun: East (sunrise), West (sunset), and from these deduced North and South.

Present the four cardinal points: North, South, East, West. Explain their clockwise arrangement (North, East, South, West) and point out that opposite directions face each other (North/South, East/West).

Organise a game in the classroom: identify together where North is (if you have a compass, use it; otherwise, base it on the position of the sun or knowledge of the neighbourhood). Place "N", "S", "E", "W" signs at the four corresponding corners of the classroom. Then ask pupils to move about: "Go towards the North!", "Turn to face the East!". This kinaesthetic game anchors the concepts in a physical way.

Show a traditional compass to the class. Let pupils observe it: what do they see? A needle that moves, letters (N, S, E, W), graduations... Explain that the needle is magnetised and always points towards magnetic North due to the Earth's magnetic field. For the more curious, you can show a short video explaining that the Earth behaves like a gigantic magnet, with magnetic North and South poles. The compass needle, which is itself magnetised, is attracted by this magnetic field.

Let pupils handle the compass (in small groups if you only have one). Ask them to verify whether the North identified earlier in the classroom is confirmed by the compass.

Announce the project: "We are going to build our own compass, but not an ordinary compass: a cardboard robot that will show us North using a miniature computer!" Show a micro:bit board and explain that it contains a special sensor called a magnetometer, capable of detecting the Earth's magnetic field, exactly like the needle of a traditional compass.

Show the LED screen of the micro:bit board: "Instead of a needle that turns, our robot will display a luminous arrow that always points North!"

Preparation and teacher notes

- Collect recycled cardboard boxes (cereal, shoe, tissue boxes) in advance – ask families a few days beforehand
- Prepare a station with cutting and gluing materials (scissors, glue, tape, felt-tip pens)
- Test access to MakeCode (makecode.microbit.org) on classroom computers and check the site is not blocked
- Charge the micro:bit boards and check the battery packs work
- Prepare an example robot template to show pupils (see appendix)

Build-up: Building and Programming the Compass Robot

Objective: Build the cardboard robot and programme the digital compass on the micro:bit board

If this is the first time pupils are using MakeCode, begin with a collective exploration of the interface projected on the screen. Identify together the different zones: the micro:bit emulator on the left, the block categories in the centre (Input, Music, LED, etc.), the programming space on the right.

Explore the "Input" category and find the "**compass heading (°)**" block. Explain that this block returns a number between 0 and 359 corresponding to the angle from magnetic North:

- **0° = North**
- **90° = East**
- **180° = South**
- **270° = West**

Depending on your pupils' level and available time, several versions of the project can be explored. Here are two versions:

PROJECT FEATURES	SIMPLE DISPLAY ON THE BOARD	DISPLAY ON THE BOARD WITH MOTORISED ARROW ON THE ROBOT
Principle	The micro:bit displays an arrow on its LEDs pointing towards North	A cardboard arrow attached to a servo motor physically rotates to point towards North
Visual Effect	Luminous arrow on the 5×5 LED screen	Mechanical arrow that actually rotates
Specific Materials	micro:bit board + battery pack only	micro:bit board + servo motor + cables + external power supply recommended
Construction Constraints	Provide a 5×4 cm opening to see the LED screen	Provide a stable location for the servo at the top of the robot + cable routing
Programming	Simple conditional blocks (if... then)	Same + angle conversion + servo control
Ideal For	First introduction to programming, large classes, limited time	Pupils comfortable with programming, in-depth project, club or workshop
Construction Tips	Create a frame with a 5×4 cm window for the LED screen. Check the board fits easily. Provide USB access at the back.	Horizontal platform at the top for the servo (vertical axis). Cardboard arrow 8–10 cm attached to the horn. Provide cable routing.



Inspiring Example from <https://www.youtube.com/watch?v=lg3VLcSRUNc>

Before building your robot, launch a collective discussion: what will our compass robot look like? Some technical constraints to respect:

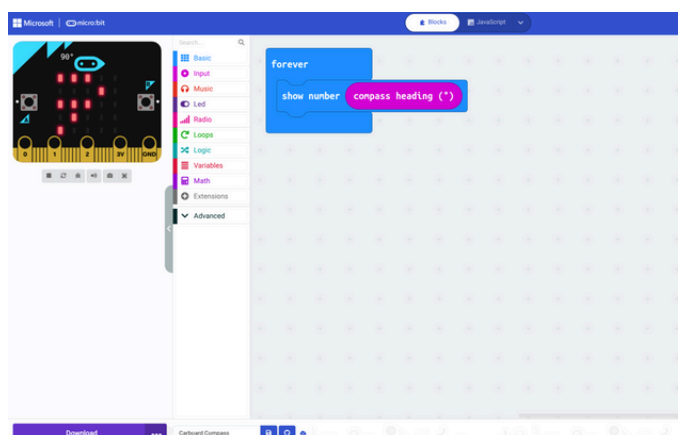
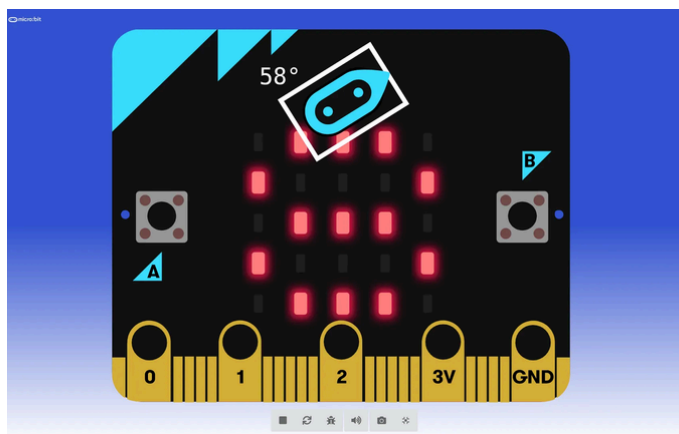
- The robot must be able to accommodate the micro:bit board visibly (the LED screen must be accessible)
- It must be stable enough to stand upright
- It must be easily transportable for orienteering activities
- The micro:bit placement must allow it to be removed and replaced easily (if shared between several robots)

Ask pupils to draw their ideal robot on paper before moving on to construction. They can draw inspiration from various shapes: humanoid robot, animal (an owl that always knows where North is), futuristic vehicle, fantastical character... then let the class in small groups build several versions. You can offer pupils the option of making the version they prefer: with or without servo motor, if you are comfortable with both solutions. Groups choose according to their preference, or you differentiate according to levels. The robots can then be compared during the final presentation.

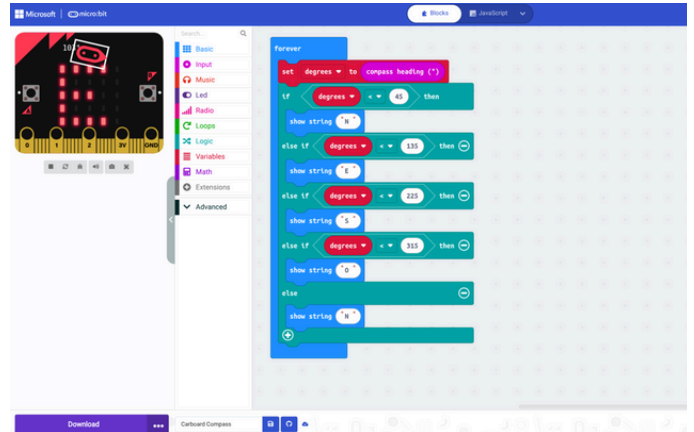
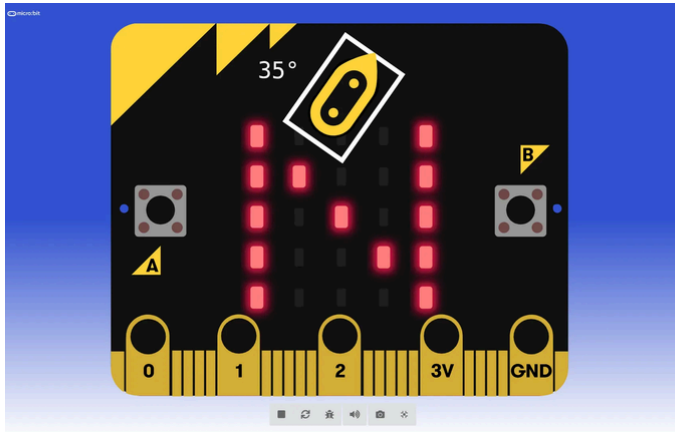
Teaching Tip: Encourage creativity whilst reminding pupils of the technical constraints. A highly decorated robot that doesn't stand up or whose LED screen can't be seen won't be functional. This is an opportunity to introduce the concept of a design brief in technology.

Once the robots are built, or in parallel to allow pupils to rotate between hands-on workshops and programming sessions, move on to creating the code that will make your "cardboard robots" work. To create the simplest code, here are the main steps – the complete code and explanation of the different blocks is available in the appendix.

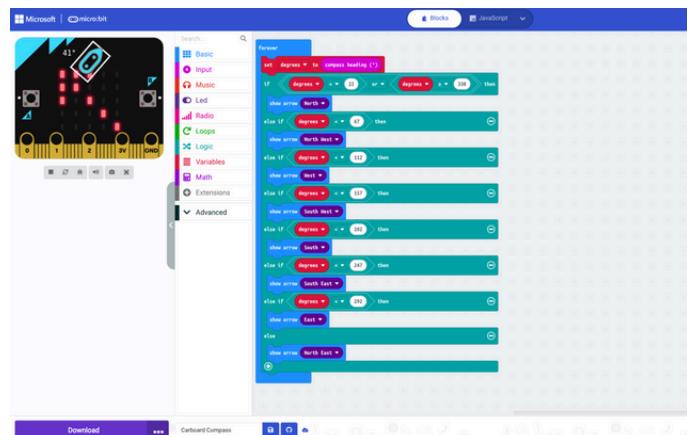
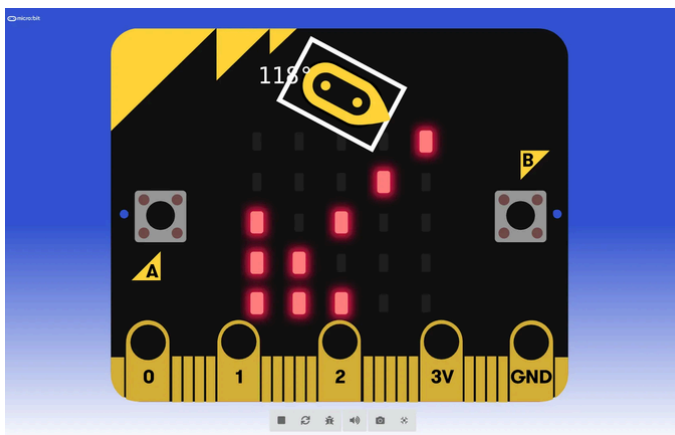
Step 1 – Display the raw value: In the "forever" block (basic block), simply display the compass direction using the "show number" block (basic block) and "compass heading (°)" block (input block). Test on the emulator (click on the simulated compass) then on a real micro:bit board. Pupils see numbers scrolling that change when they rotate the board.



Step 2 – Display the cardinal point letter: Use "if... then... else" blocks to display N, E, S, or W depending on the angle value. For example: if the angle is between 315° and 45°, display "N"; if between 45° and 135°, display "E", etc.

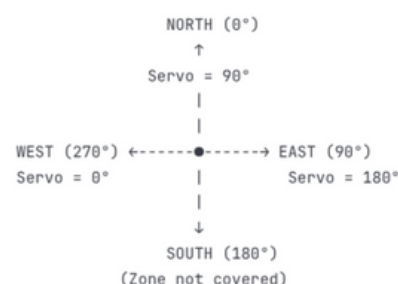


Step 3 – Display an arrow pointing towards North: Use LED blocks to create arrows in all 8 directions. The arrow always points towards North, so if the user faces North (0°), the arrow points upward; if they face East (90°), the arrow points left (because North is then to their left). At this stage, you can also add more detail, specifying North-West, South-West, etc.



For groups that have chosen the version with servo motor, continue with these additional steps:

Step 4 - Understand the principle: The servo motor can only rotate from 0° to 180°, i.e., a half-turn. So we will use the arrow to indicate North only on the front arc: from West (to the left) to East (to the right), passing through North (in front). When North is behind us (in the South zone), the arrow will stay blocked on one side to indicate "North is that way, turn around!". Here is the correspondence diagram between the compass and servo motor:

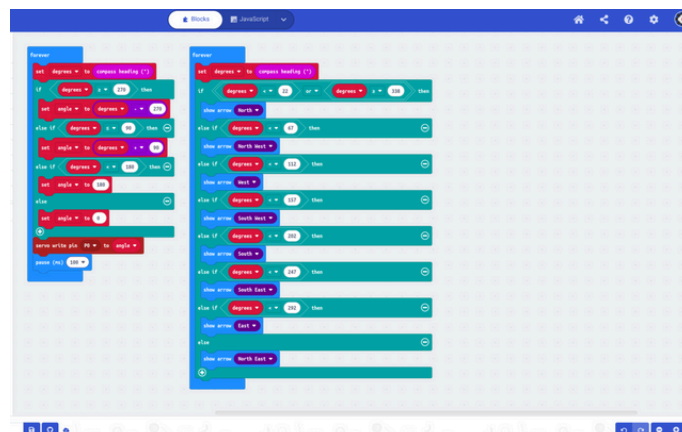
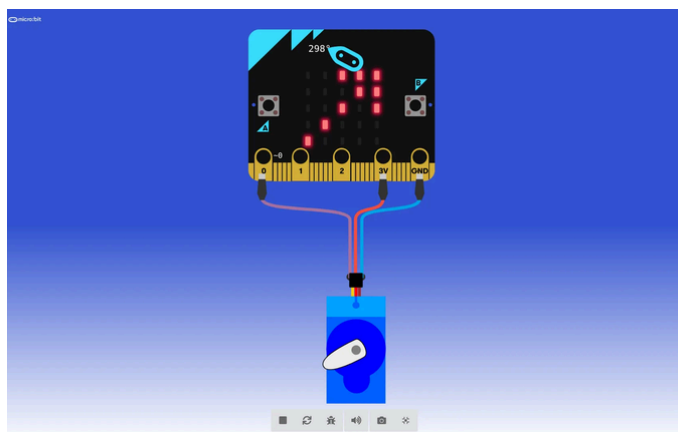


COMPASS	→	SERVO MOTOR
270° (West)	→	0° (left)
315° (North-West)	→	45°
0° (North)	→	90° (front)
45° (North-East)	→	135°
90° (East)	→	180° (right)
91° to 269° (South)→ Blocked at edges		

Step 5 - Create variables and calculate: In the "Variables" category, create a new variable "angle" (for the servo position). The variable "degrees" created in step 2 already contains the compass direction. Use "if... then... else" blocks (Logic category) with calculation blocks (Maths category) to calculate the servo angle according to the diagram above:

- If "degrees" is greater than or equal to 270 (West to North zone), then "set angle to degrees - 270"
- Otherwise, if "degrees" is less than or equal to 90 (North to East zone), then "set angle to degrees + 90"
- Otherwise (we are in the South zone, between 91° and 269°), block the arrow at the nearest edge: if "degrees" is less than 180, then "set angle to 180" (blocked on the East side), otherwise "set angle to 0" (blocked on the West side)

Step 6 - Control the servo motor: Click on "Advanced" to reveal the "Pins" category. Use the "servo write pin P0 to" block and place the "angle" variable in it. Add a 100 ms pause (Basic category, "pause (ms)" block) to stabilise the movement. Test: when you turn to face North, the arrow points forward; facing East, it points right; facing West, it points left.





If you have few physical micro:bit boards, organise rotating workshops: a "Programming" workshop where pupils code on the MakeCode emulator, a "Construction" workshop where they build and decorate their robot, and a "Real Testing" workshop where a small group uses the physical boards to test, calibrate, and validate. Rotate all groups through these three workshops.

Mathematical Extension – The Symmetry of Cardinal Points

Take advantage of this activity to get pupils thinking about the symmetry of directions. If the arrow shows where North is, where is South? Answer: always opposite, 180° apart! Pose riddles: "The arrow points to the right, so North is East. Where is South?" (West, behind to the left). "The arrow points forward, where is South?" (Behind us). This reflection helps understand that with just North, we can deduce all other directions: South is opposite (+180°), East is 90° clockwise, West is 90° anticlockwise. This is also why our servo covering only 180° is sufficient: knowing North, we know everything!

Calibration: The first time you use the compass on a real micro:bit board, it requires calibration. You must tilt the board in all directions until all the LEDs light up, forming a circle. This is an essential step for accurate measurements. Allow time for this step with pupils.



Preparation and teacher notes

- Scout the terrain in advance: identify suitable areas for courses (schoolyard, nearby park) and check exit permissions if necessary
- Prepare and print maps of the location with North indicated (use Ordnance Survey maps or OpenStreetMap)
- Set up markers or coloured cones before the session for orienteering courses
- Write the directional clues for the treasure hunt and hide them at the planned locations
- Provide logbooks or sheets for pupils to record their directions

Rehearsal: Using the Compass Robot to Orient Oneself

Objective: Put orientation skills into practice with the compass robot through fun activities

Distribute to each group a map of the schoolyard or nearby neighbourhood.

The first task is to learn to "orient" the map, i.e., position it so that the North on the map corresponds to real North:

1. Lay the map flat in front of you
2. Hold the compass robot and observe the arrow: it points towards North
3. Rotate the map until the "N" arrow on the map is aligned with the direction shown by the robot
4. The map is now oriented: what is on the right on the map is actually on your right!

Ask pupils to identify on the oriented map where various elements are located: the school building, the gate, the covered area, the trees... Check together that everything matches reality.

Prepare a course in the schoolyard or a space near the school. At each stage, pupils find a clue giving them a direction to follow (example: "Go East for 20 paces"). The compass robot helps them find the right direction.

Example clues:

- ***"Walk North until you reach the big tree"***
- ***"Turn to face West and take 15 paces"***
- ***"The next clue is to the South-East, near the bench"***

Pupils use their compass robot to orient themselves and follow the directions. They record their route in their logbook, tracing the directions followed.

For a more sporting version, organise a proper orienteering race in a nearby park (with necessary permissions). Pupils must find several markers in a set order, using a map and their compass robot:

- Place 5 to 8 markers (cones, flags) at different locations
- Mark their position on the map with numbers
- Teams set off at regular intervals
- At each marker, pupils collect a stamp or note a code
- The team that completes the course fastest with all codes wins

Adaptation: If an outdoor trip is not possible, create an orienteering course within the school itself: corridors, classrooms, playground. The learning objective remains the same, only the space changes.

Preparation and teacher notes

- Prepare an exhibition space for pupils' work
- Inform parents
- Keep photos to document the work done

Clean-up: Present and Reflect on Learning

Objective: Showcase the work accomplished and reflect on the learning achieved throughout the project

Once all activities are complete, you can organise an exhibition of the compass robots in the classroom or in a communal area of the school. Each group presents their robot: its name, its design, the programme used, and gives a demonstration of how it works. Invite other classes, parents, or school representatives to come and discover the creations. Pupils explain how the compass works, the cardinal points, and show how to orient a map.

Conclude the project with a moment of collective reflection. Ask open questions to pupils:

- "What did you learn about orientation that you didn't know before?"
- "How does a compass work? Explain it in your own words."
- "What was the most difficult part of the project? The most fun?"
- "In what everyday situations could you use these orientation skills?"



Help pupils become aware of the skills developed: reading a map, using cardinal points, programming, working as a team, solving technical problems...

Create together a written summary of the project, including:

- A diagram of the cardinal points with degree values
- An explanation of how a compass works (terrestrial magnetism)
- The stages of robot construction
- The code used (in blocks or simplified pseudocode)
- Photos of the project and orienteering activities

Assessment Approaches - Some hints

Assessment of this project can take several complementary forms:

- **Ongoing Assessment:** Observe skills in action during the different phases (ability to orient oneself, read a map, programme, collaborate).
 - **Assessable Productions:** The functional robot, the programme created, participation in orienteering courses, the oral presentation.
 - **Self-Assessment:** Have pupils complete a grid: "What I learnt", "What I succeeded at", "What was difficult"
- 

Appendix: Example of a ready-to-use Micro:bit Program

If you're using a micro:bit for the first time

- Program using the MakeCode editor: Go to makecode.microbit.org to program your board directly in the browser
- Connect the board to the computer: Use a micro-USB cable. The board will appear as a USB drive named "MICROBIT". Your board will be automatically associated with your editor. If not, you can manually associate it by clicking the "Connect device" button in the menu available by clicking "..." next to the "Download" button.
- Transfer the program: Once your code is created, click "Download" and the program will automatically download to the associated board. If not, you can copy the .hex file to the board. The orange LED blinks during transfer.
- Autonomous power: To collect outdoor data, add an AAA battery pack (not included with the board) in the provided holder.
- Data recovery: Reconnect the board to the computer, open MakeCode, and click "Show data" to download the recorded measurements in CSV format.

For more information: [micro:bit - Getting Started - Introduction](#)

Sample Code for Version 1 - Without Servo Motor

```
let degrees = 0
basic.forever(function () {
  degrees = input.compassHeading()
  if (degrees < 22 || degrees >= 338) {
    basic.showArrow(ArrowNames.North)
  } else if (degrees < 67) {
    basic.showArrow(ArrowNames.NorthWest)
  } else if (degrees < 112) {
    basic.showArrow(ArrowNames.West)
  } else if (degrees < 157) {
    basic.showArrow(ArrowNames.SouthWest)
  } else if (degrees < 202) {
    basic.showArrow(ArrowNames.South)
  } else if (degrees < 247) {
    basic.showArrow(ArrowNames.SouthEast)
  } else if (degrees < 292) {
    basic.showArrow(ArrowNames.East)
  } else {
    basic.showArrow(ArrowNames.NorthEast)
  }
})
```

Do not hesitate to copy paste for testing.

Access the code online: <https://makecode.microbit.org/S44089-00150-97024-09020>

Appendix: Example of a ready-to-use Micro:bit Program

What does the programme do?

The programme continuously reads the compass direction (in degrees from 0 to 359). Depending on the angle measured, it displays one of 8 possible arrow directions. Note: the arrow indicates where North is, not where the user is looking! For example, if the user faces East (90°), North is to their left, so the arrow points left (West on the screen).

Direction	Degrees
North	338° – 22°
North-East	22° – 67°
East	67° – 112°
South-East	112° – 157°
South	157° – 202°
South-West	202° – 247°
West	247° – 292°
North-West	292° – 338°

Appendix: Example of a ready-to-use Micro:bit Program

Sample Code for Version 2 - With Servo Motor

```
let degrees = 0
let angle = 0
basic.forever(function () {
    degrees = input.compassHeading()
    if (degrees >= 270) {
        angle = degrees - 270
    } else if (degrees <= 90) {
        angle = degrees + 90
    } else if (degrees < 180) {
        angle = 180
    } else {
        angle = 0
    }
    pins.servoWritePin(AnalogPin.P0, angle)
    basic.pause(100)
})
basic.forever(function () {
    degrees = input.compassHeading()
    if (degrees < 22 || degrees >= 338) {
        basic.showArrow(ArrowNames.North)
    } else if (degrees < 67) {
        basic.showArrow(ArrowNames.NorthWest)
    } else if (degrees < 112) {
        basic.showArrow(ArrowNames.West)
    } else if (degrees < 157) {
        basic.showArrow(ArrowNames.SouthWest)
    } else if (degrees < 202) {
        basic.showArrow(ArrowNames.South)
    } else if (degrees < 247) {
        basic.showArrow(ArrowNames.SouthEast)
    } else if (degrees < 292) {
        basic.showArrow(ArrowNames.East)
    } else {
        basic.showArrow(ArrowNames.NorthEast)
    }
})
```

Do not hesitate to copy paste for testing.

Access the code online: <https://makecode.microbit.org/S00541-68289-24489-33866>

Appendix: Example of a ready-to-use Micro:bit Program

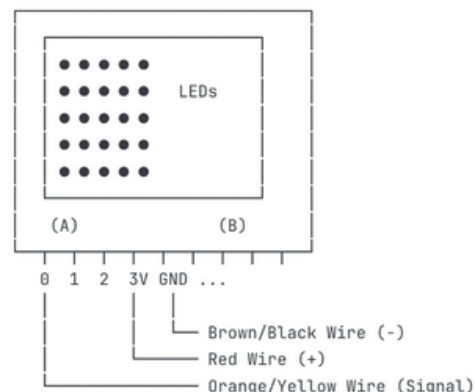
What does the programme do and how to connect the servo motor?

This programme rotates a mechanical arrow attached to a servo motor to point towards North. The servo motor has three wires to connect to the micro:bit board:

Servo Wire	Common Colour	micro:bit Pin
Signal	Orange or Yellow	P0
Power +	Red	3V
Ground -	Brown or Black	GND

Note: The servo motor draws current. For prolonged use, it is preferable to power the servo with an external source (pack of 4 AA batteries) rather than directly from the micro:bit board. For short demonstrations, power from the board works fine. Also avoid extreme angles (0° and 180°) which make the servo grind; prefer a range of 10° to 170° .

Wiring diagram on the micro:bit board



Conversion Formulae:

Compass Zone	Condition	Servo Angle Calculation
West to North	degrees ≥ 270	angle = degrees - 270
North to East	degrees ≤ 90	angle = degrees + 90
South-East (blocked)	$90 < \text{degrees} < 180$	angle = 180
South-West (blocked)	$180 \leq \text{degrees} < 270$	angle = 0

Concrete Examples:

Compass	Calculation	Servo	Arrow Position
270° (West)	$270 - 270$	0°	Points left
315° (North-West)	$315 - 270$	45°	Points front-left
0° (North)	$0 + 90$	90°	Points forward
45° (North-East)	$45 + 90$	135°	Points front-right
90° (East)	$90 + 90$	180°	Points right
135° (South-East)	Blocked	180°	Stays right
225° (South-West)	Blocked	0°	Stays left